

CURIE-OSITY

Curie-osity, a journey where science goes beyond formulas, facts, and textbooks. This category is all about how well you can think, question, reason, and adapt when the answer isn't obvious. Every challenge will push you to look beyond what's in front of you, connect scientific ideas, and trust your logic under pressure. Expect uncertainty, unexpected turns, and moments that make you question what you thought you knew. Knowledge may give you a head start, but curiosity, reasoning, and instinct will take you further.

ROUND 1 : Spot the flaw

ROUND DURATION: 3 HOURS

DELEGATE CAP: 2-3

Not everything presented as science is necessarily correct. In Spot the Flaw, delegates will encounter a range of scientific material from Biology, Chemistry and Physics, where careful observation can be just as important as subject knowledge.

Delegates will work together to examine the information placed before them and decide what does and does not make scientific sense. The challenges may involve experimental situations, results, explanations or applications of familiar scientific concepts. Simply spotting something unusual will not always be enough; delegates should be able to support their judgement using relevant scientific reasoning.

Physics, Chemistry and Biology can all come up in this round, so delegates should come prepared for anything. The main thing is to think critically, question what's being presented to them, and most importantly, never just assume that what's in front of them is correct.

Judging Criteria:

Cases with a Flaw:

1. Correct Verdict: 2 pts

Correctly deciding whether the case has a significant scientific flaw based on the information and evidence given.

2. Identification of Main Flaw: 4 pts

Identifying the actual scientific flaw in the case. This includes pointing out the main problem in the method, evidence, investigation, reasoning or conclusion, rather than focusing on small limitations that do not really affect the result.

3. Scientific Reasoning & Justification: 5 pts

Explaining clearly why the identified flaw is a problem using correct scientific concepts and evidence from the case. The reasoning should show how the flaw affects the reliability, validity or conclusion of the investigation.

4. Correction: 2 pts

Giving a suitable scientific correction for the flaw identified. The correction should directly solve the problem and explain how the investigation or conclusion could be improved.

Cases without a Flaw:

1. Correct Verdict: 2 pts

Correctly recognising that there is no significant scientific flaw in the case. Minor limitations or unfamiliar details should not be treated as major errors.

2. Scientific Reasoning & Justification: 3 pts

Explaining why the case is scientifically sound using the information and evidence given. The reasoning should clearly defend why the method, evidence and conclusion are acceptable despite any minor limitations.

What to Prepare :

- Application of Biology, Chemistry and Physics
- Basic experimental methods
- Understanding of scientific data and results
- Identifying errors or flaws in scientific investigations
- Scientific reasoning and justification
- Suggesting improvements to an investigation
- Being able to defend a scientifically correct case

Note:

Devices are not allowed in this round.

This is an elimination round. Only 26 teams will advance.

ROUND 2: False Positive

DELEGATE CAP: 1

Round 2 takes delegates into a challenge where not everything can be taken at face value. Scientific knowledge from Biology, Chemistry and Physics will be important, but the way delegates are expected to use it will only be revealed when the round begins.

Delegates should be prepared to think on the spot, question what they hear, and make careful decisions under pressure.

The exact format and mechanics of this round will remain undisclosed until the event.

(Hint: Delegates should have vast knowledge of apparatuses and chemicals used in these disciplines.)

Note:

Devices are not allowed in this round.

This is an elimination round. Only 10 teams shall advance to the next round.

ROUND 3 : Blueprint Unknown

ROUND DURATION: 4 Hours

DELEGATE CAP: 3-4

Teams will be given a hands-on construction challenge using materials provided during the round. The task will require delegates to plan, build, and apply basic scientific and engineering principles to create a structure that meets the given requirements.

However, building alone will not be enough. Scientific questions will be introduced during the challenge, and incorrect answers may make the final test harder for the team. Delegates will therefore need to balance their construction with their knowledge of Biology, Chemistry and Physics.

The exact build and testing conditions will remain undisclosed until the round begins.

Note:

This is an elimination round. Only 6 teams will advance.

ROUND 4: Case in Question

DELEGATE CAP: 3

Teams will be given a science-based case or motion on the spot and will choose whether to stand for or against it. Topics may involve scientific developments, real-world issues, or situations where science and ethics overlap.

Delegates will be expected to form a well-supported stance using scientific reasoning, evidence, and relevant examples. They should also be prepared to deliver an opening statement, argument, rebuttal, and closing statement while defending their position against the opposing side.

1. Scientific Accuracy: 17 pts

How scientifically correct the facts, concepts and explanations used by the team are throughout the debate.

2. Strength of Argument: 16 pts

How logically and convincingly the team builds its assigned stance. Arguments should be relevant, well-developed and connected to the case.

3. Use of Evidence: 14 pts

How well the team supports its claims using information from the case and relevant scientific knowledge rather than unsupported statements.

4. Rebuttal: 16 pts

How effectively the team responds to the opposing side by identifying weaknesses, incorrect scientific claims or unsupported assumptions and directly challenging them.

5. Clarity: 7 pts

How clear, organised and easy to understand the team's arguments and scientific explanations are.

6. Team Coordination: 5 pts

How smoothly delegates divide their roles, transition between speakers and support each other throughout the debate.